Experimental Setup and Design

A. Experimental Setup

1. Workload Input

- Batch jobs extracted from:
  - Alibaba Cluster Trace
  - Azure VM/Function traces
- Extracted parameters:
  - Job arrival time
  - CPU and memory requirement
  - Execution duration
  - Deadline/slack window

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2. Carbon Data Integration

- Hourly carbon intensity values
- Region-specific grid emission signal
- 24-hour rolling forecast window
- Time-aligned with workload traces

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3. Pricing Model Integration

- AWS Spot price history
- Reserved instance cost modeling
- Idle cost calculation
- SLA violation penalty modeling

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4. Simulation Environment

- Discrete time-slot simulation (e.g., 1-hour intervals)
- Hybrid cloud model:
  - Reserved instances
  - Spot instances
- Probabilistic job duration handling

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5. Baseline Algorithms for Comparison

To prove your method works, compare against:

1. Immediate Execution (No Shifting)
2. Static Carbon Threshold Policy
3. Greedy Carbon-Minimization Policy
4. Cost-Only Optimization Policy

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6. Evaluation Metrics

Environmental Metrics:

- Total carbon emissions (kg CO_2)
- % Carbon reduction vs baseline

Economic Metrics:

- Total operational cost
- Reserved instance idle cost
- Spot instance utilization

Performance Metrics:

- Throughput
- Average job delay
- SLA violation rate

B. Experimental Design Strategy

1. Trade-Off Analysis

- Vary weight parameters (α, β, γ)
- Plot Carbon vs Cost vs SLA trade-off curves

2. Sensitivity Analysis

- Vary workload intensity (low, medium, high load)
- Vary carbon volatility (stable vs highly fluctuating grid)

3. Robustness Testing

- Test under:
  - Highly unpredictable job durations
  - Sudden carbon spikes
  - Spot price volatility

4. Scalability Testing

- Increase number of jobs
- Increase number of machines
- Measure algorithm runtime overhead

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3) Equipment and Software Used

A. Hardware Equipment

- Personal workstation / research lab server
  - Intel i5/i7 or equivalent processor
  - Minimum 16 GB RAM
  - 512 GB SSD
- Optional: Cloud VM for large-scale simulation
- Stable internet connection (for dataset download)

B. Software and Tools

1. Programming Environment

- Python 3.9+
- Jupyter Notebook / Google Colab

2. Libraries

- NumPy (Numerical operations)
- Pandas (Dataset preprocessing)
- Matplotlib (Graph plotting)
- SciPy (Optimization support)
- Scikit-learn (Probabilistic modeling)

3. Cloud & Data Tools

- AWS CLI (Spot price extraction)
- Electricity Maps API
- WattTime API
- Cloud Carbon Footprint tool

4. Development Environment

- VS Code / PyCharm
- GitHub (Version control)
- LaTeX (IEEE paper formatting)
- PowerPoint (Presentation preparation)